

AI-BASED RESUME ANALYZER FOR SKILL EXTRACTION AND JOB ROLE PREDICTION USING NLP

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ABSTRACT

In the modern recruitment process, evaluating large volumes of resumes manually is time-consuming, inefficient, and often prone to human bias. This paper presents an AI-based Resume Analyzer system designed to automate the process of resume screening and candidate evaluation. The system leverages Natural Language Processing (NLP) techniques to extract key information such as skills, education, experience, and keywords from resumes. It then matches the extracted data with predefined job role requirements to generate a suitability score for each candidate. The proposed system improves the accuracy and speed of resume evaluation by reducing manual effort and ensuring fair and consistent candidate assessment. Additionally, it provides insights such as skill gaps and job role recommendations, helping candidates understand their suitability for specific positions. The system is implemented using Python and integrates machine learning-based text processing techniques for efficient analysis. Experimental results show that the system effectively classifies resumes based on relevance to targeted job roles and provides meaningful evaluation metrics. This approach demonstrates the potential of AI-driven tools in enhancing recruitment processes and improving decision-making efficiency in human resource management.

Keywords: Resume Analysis, Natural Language Processing, Machine Learning, Artificial Intelligence, Skill Extraction, Job Role Matching, Automated Recruitment, Text Mining.

1. INTRODUCTION

The recruitment process plays a crucial role in selecting suitable candidates for various job roles in an organization. Traditionally, this process involves manually screening resumes, shortlisting candidates, and evaluating their skills based on job requirements. However, with the rapid increase in the number of applicants for each position, manual resume screening has become time-consuming, inefficient, and susceptible to human bias and errors.

To address these challenges, automation in recruitment using Artificial Intelligence (AI) and Natural Language Processing (NLP) has gained significant attention. AI-based resume analysis systems can efficiently process large volumes of resumes, extract meaningful information, and match candidate profiles with job role requirements. This not only speeds up the hiring process but also improves accuracy and consistency in candidate evaluation.

In this paper, we propose an AI-based Resume Analyzer system that automatically processes resumes and extracts key information such as skills, education, experience, and keywords using NLP techniques. The extracted data is then compared with predefined job role criteria to generate a suitability score for each candidate. The system also provides recommendations based on skill matching, helping both recruiters and candidates make better decisions.

The objective of this system is to reduce manual effort, eliminate bias, and improve the efficiency of the recruitment process. By leveraging machine learning and text processing techniques, the proposed solution aims to enhance the overall hiring workflow and support intelligent decision-making in human resource management.

2. LITERATURE REVIEW

Several research studies have been conducted in the field of resume analysis and intelligent recruitment systems using Artificial Intelligence (AI) and Natural Language Processing (NLP). These studies aim to improve the efficiency, accuracy, and fairness of the hiring process by automating resume screening and candidate evaluation.

Lakshmi et al. [1] proposed an automated resume screening system that uses NLP techniques to extract relevant information such as skills, education, and work experience from resumes for intelligent talent shortlisting. Their

approach demonstrated that automation significantly reduces the time required for initial candidate shortlisting compared to manual methods.

Murthy et al. [2] developed a DistilBERT-based NLP framework that classifies resumes into different job categories based on automated skill extraction. The study highlighted that transformer-based NLP systems improve classification accuracy and reduce human bias in the selection process.

Jain et al. [3] introduced an intelligent resume screening system using machine learning and NLP techniques to rank candidates and identify the most suitable profiles for specific job roles. Their results showed improved efficiency in the overall screening process.

Zaroor et al. [4] presented a job post and resume classification system (JRC) for online recruitment, focusing on matching candidate profiles with job requirements. Their system effectively identified core competencies and matched them with job requirements, improving decision-making in recruitment systems.

Overall, existing literature indicates that AI and NLP-based resume screening systems significantly enhance recruitment efficiency by automating manual tasks, reducing bias, and improving candidate-job matching accuracy. However, many existing systems still face challenges such as handling unstructured resume formats and improving semantic understanding of skills, which are addressed in the proposed system.

3. METHODOLOGY

The proposed AI-based Resume Analyzer system is designed to automate the process of resume screening and candidate evaluation using Natural Language Processing (NLP) and Machine Learning (ML) techniques. The methodology focuses on extracting meaningful information from resumes, analyzing it, and matching it with predefined job role requirements to generate a suitability score.

3.1 Data Collection

The system accepts resumes in common formats such as PDF and DOCX. Additionally, job descriptions for different roles are defined as input criteria. These job descriptions act as a reference dataset for matching candidate profiles.

3.2 Data Preprocessing

The extracted resume text is cleaned and preprocessed to improve analysis accuracy. This step includes:

- Removal of special characters, symbols, and stop words
- Tokenization of text into meaningful words
- Conversion to lowercase for uniformity
- Lemmatization to reduce words to their base form

This ensures that only relevant textual information is used for further processing.

3.3 Feature Extraction

After preprocessing, key features are extracted from resumes using NLP techniques. These features include:

- Skills (technical and non-technical)
- Educational qualifications
- Work experience
- Keywords relevant to job roles

Techniques such as TF-IDF (Term Frequency–Inverse Document Frequency) or vectorization are used to represent textual data in numerical form for machine learning models.

3.4 Job Matching and Scoring

The extracted resume features are compared with predefined job role requirements. A similarity score is calculated based on keyword matching and semantic relevance. Machine learning algorithms or cosine similarity measures are used to evaluate how closely a candidate matches a specific job role.

3.5 Classification and Recommendation

Based on the similarity score, candidates are classified into categories such as:

- Highly Suitable

- Moderately Suitable
- Not Suitable

The system also provides job role recommendations based on the strongest matching skills identified in the resume.

3.6 Output Generation

Finally, the system generates a structured output that includes:

- Resume evaluation score
- Extracted skills and experience summary
- Recommended job roles
- Skill gap analysis (if applicable)

3.7 Tools and Technologies

The system is implemented using the following tools and technologies:

- Python as the primary programming language
- Natural Language Processing (NLTK / spaCy) for text preprocessing and feature extraction
- Scikit-learn for similarity computation and classification
- Streamlit for building the user interface
- Google Gemini API for generating AI-powered resume insights, candidate summaries, and intelligent job-role recommendations

The integration of the Gemini API enhances the system by providing contextual understanding of resumes and generating human-like analytical feedback, improving the overall decision-support capability of the system.

3.8 Formulas Used in the System

1) Cosine Similarity, measuring similarity between the resume vector A and job description vector B:

$$\text{Similarity} = \cos(\theta) = (A \cdot B) / (|A| |B|) \quad (1)$$

$A \cdot B$ is the dot product and $|A|$, $|B|$ are vector magnitudes, yielding a score in $[0,1]$ that is scaled to a percentage.

2) Resume Score, obtained by scaling the similarity score, or in a weighted configuration:

$$\text{Resume Score} = \text{Similarity Score} \times 100 \quad (2)$$

$$\text{Resume Score} = (W_s \cdot S) + (W_k \cdot K) + (W_e \cdot E) \quad (3)$$

where S, K, E are the skill, keyword, and experience match scores, and W_s , W_k , W_e are their respective weights.

3) Skill Matching Ratio:

$$\text{Skill Match Ratio} = (\text{Matched Skills} / \text{Total Required Skills}) \times 100 \quad (4)$$

4) Skill Gap, the number of missing skills for the target role:

$$\text{Skill Gap} = \text{Required Skills} - \text{Matched Skills} \quad (5)$$

TF-IDF/embedding vectorization builds the resume and job-description vectors, the Gemini API improves semantic matching beyond keyword overlap, and the scaled cosine similarity produces the score shown to the user. For example, a Data Scientist candidate with 7 of 8 required skills has a skill gap of $8 - 7 = 1$, consistent with Fig. 4.

The proposed methodology efficiently automates resume analysis by combining NLP-based text processing and machine learning-based matching techniques, thereby improving recruitment speed, accuracy, and fairness.

4. SYSTEM ARCHITECTURE

The architecture of the proposed system is modular, with interconnected components that process resumes, extract information, and generate job-role matching results, as illustrated in Fig. 1.

4.1 Overview of Architecture

The system follows a pipeline-based architecture, where each stage handles a specific task—input, preprocessing, feature extraction, or classification—with data flowing sequentially from upload to final output.

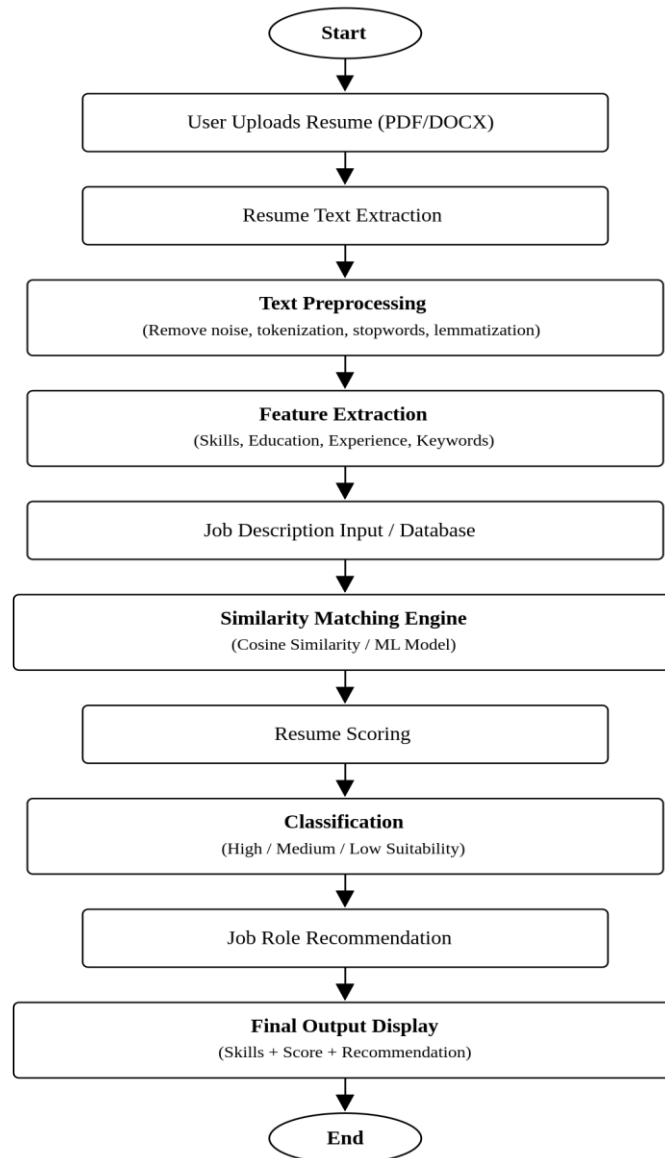


Fig. 1. Workflow of the proposed AI-based Resume Analyzer.

4.2 Components of System Architecture

- 1) User Interface (UI): Built using Streamlit, the UI is the system's entry point, where users upload resumes in PDF or DOCX format.
- 2) Resume Input Module: Converts the uploaded file into machine-readable text using libraries such as PyPDF2 or docx2txt.
- 3) Text Preprocessing Module: The extracted text is cleaned and normalized. This includes:
 - Removing special characters and punctuation
 - Tokenization of text
 - Stop word removal
 - Lemmatization

This step ensures that only relevant and structured data is passed to the next stage.

- 4) Feature Extraction Module: Extracts skills, education, experience, and keywords from the preprocessed text, using TF-IDF or word embeddings (Word2Vec / BERT) to convert text into numerical vectors.

- 5) Job Description Matching Engine: Compares extracted resume features with predefined job descriptions using cosine similarity or ML-based classification to determine job-role fit.

6) Scoring and Classification Module: Based on similarity results, the system assigns a resume score and classifies candidates into categories such as:

- Highly Suitable
- Moderately Suitable
- Not Suitable

7) Output Module: The final module presents results to the user. It displays:

- Resume score
- Extracted skills and profile summary
- Recommended job roles
- Skill gap analysis (if applicable)

4.3 Data Flow Summary

Resume input → Text extraction → Preprocessing → Feature extraction → Matching engine → Scoring → Output display.

The proposed architecture ensures an efficient and scalable resume analysis system by breaking down the process into independent modules. This modular design improves maintainability, accuracy, and flexibility, making it suitable for real-world recruitment applications.

5. RESULTS

The proposed AI-based Resume Analyzer was implemented and tested using sample resumes across different job roles, processing PDF and DOCX formats and evaluating candidates against predefined job descriptions.

The system was evaluated on a sample dataset of 10 resumes tested against five target job roles—Data Scientist, Python Developer, Data Analyst, Machine Learning Engineer, and DevOps Engineer—collected from publicly available sources such as Kaggle and self-prepared resumes. Each resume was tested against its corresponding job description to assess skill extraction, similarity scoring, and suitability classification.

5.1 Resume Parsing and Feature Extraction Results

The system was able to accurately extract key information such as skills, educational qualifications, and work experience from resumes. The NLP-based preprocessing techniques improved text clarity by removing noise and irrelevant data. It was observed that structured resumes produced higher extraction accuracy compared to unstructured formats. Fig. 2 shows the resume upload interface, where the user selects the target job role before analysis.

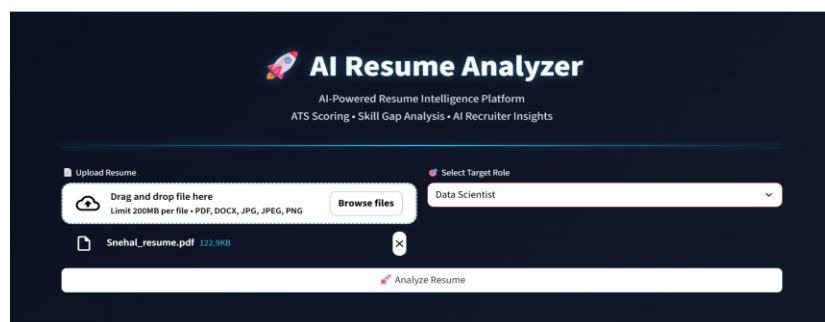


Fig. 2. Resume upload interface with target job role selection.



Fig. 3. Resume score, career match, skill count, and readiness output.

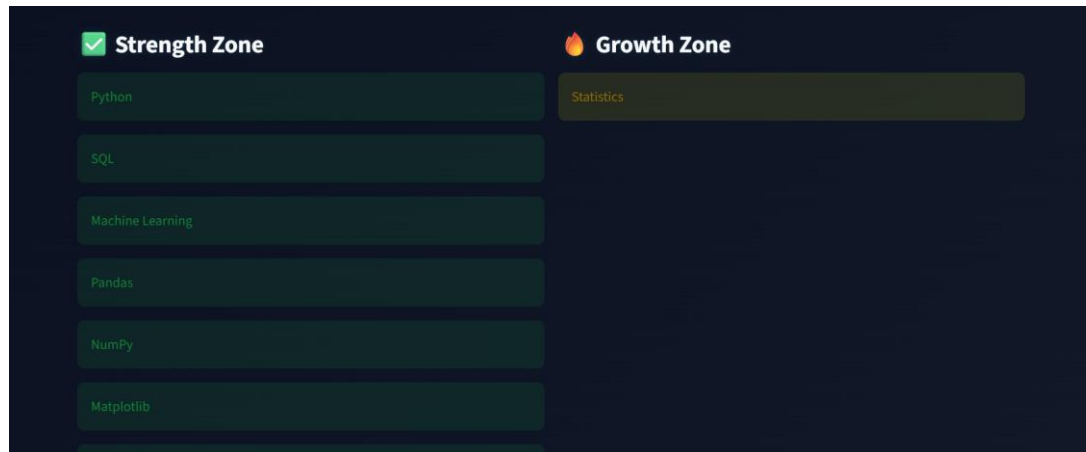


Fig. 4. Strength Zone (matched skills) and Growth Zone (missing skills) for the target role.

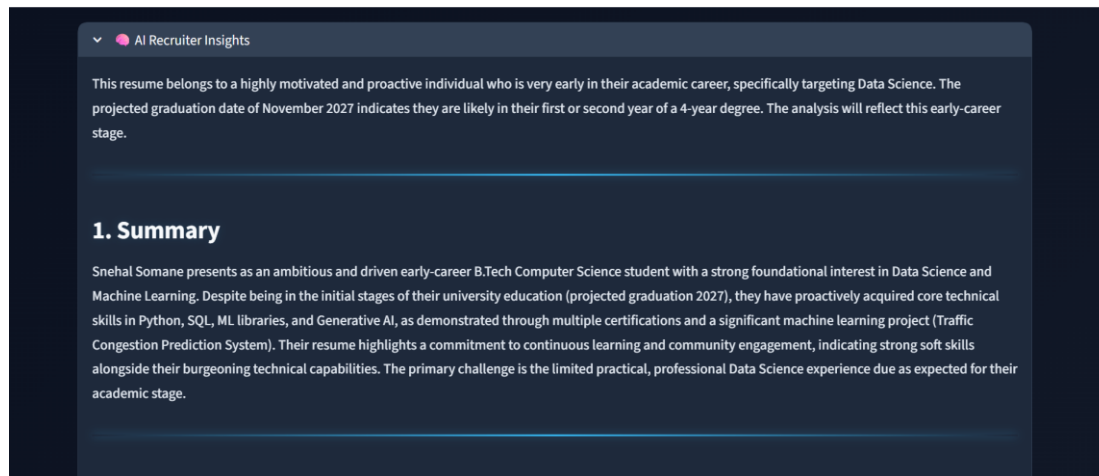


Fig. 5. AI-generated recruiter insight summary of the candidate profile.

Table 1. Aggregated Resume Evaluation Results by Job Role

Job Role	Resumes Tested	Avg. Score (%)	Avg. Skills Matched	Avg. Skills Missing	Suitability Mix
Data Scientist	3	66.3	5.3	2.7	1H, 1M, 1N
Python Developer	2	37.0	3.0	5.0	2N
Data Analyst	3	47.0	3.3	4.3	1H, 2N
Machine Learning Engineer	1	28.0	2.0	6.0	1N
DevOps Engineer	1	75.0	6.0	2.0	1M

H = Highly Suitable, M = Moderately Suitable, N = Not Suitable. Values are averaged across all resumes tested for each role.

Table 2. Summary of Results

Metric	Value
Total Resumes Tested	10
Highest Score Obtained	87%

Lowest Score Obtained	28%
Highly Suitable Candidates	2
Moderately Suitable Candidates	2
Not Suitable Candidates	6

As shown in Fig. 3, the system achieved a resume score of 87% for a sample candidate profile targeting the Data Scientist role, with 7 relevant skills identified and an overall readiness rating of “High.”

5.2 Job Matching Performance

The similarity matching module effectively compared extracted resume features with job role requirements. Using cosine similarity-based scoring, the system successfully identified the relevance of candidates to specific job roles. Candidates with matching skill sets received higher similarity scores, indicating better suitability. Fig. 4 illustrates the Strength Zone and Growth Zone breakdown generated for a sample candidate, showing matched skills (e.g., Python, SQL, Machine Learning, Pandas, NumPy, Matplotlib) against the single missing skill (Statistics) for the target role.

5.3 Classification Results

Based on computed similarity scores, candidates were classified into three categories:

- Highly Suitable
- Moderately Suitable
- Not Suitable

The classification results showed that the system consistently ranked candidates according to their skill relevance and alignment with job descriptions. Table 1 presents the average resume score, average matched and missing skills, and overall suitability mix aggregated by job role across all 10 test resumes. Table 2 summarizes the results at the individual-candidate level: scores ranged from 28% to 87%, with 2 candidates rated Highly Suitable, 2 Moderately Suitable, and 6 Not Suitable.

5.4 Overall System Performance

Across all 10 test resumes, scores ranged from 28% to 87%, with 2 candidates classified as Highly Suitable, 2 as Moderately Suitable, and 6 as Not Suitable, as summarized in Table 2. The system demonstrated efficient performance in terms of:

- Reduced manual screening time
- Improved accuracy in candidate evaluation
- Consistent and unbiased scoring mechanism
- Faster processing of multiple resumes

The results indicate that the proposed system significantly enhances the recruitment process by automating resume screening and improving decision-making efficiency.

5.5 Observations

It was observed that the accuracy of the system depends on the quality of resume formatting and the completeness of job descriptions. The system performs best when resumes contain clearly defined skills and structured information. In addition to numerical scoring, the system also generates a natural-language recruiter insight summary, as shown in Fig. 5, which qualitatively describes the candidate's profile, strengths, and areas for development.

Notably, scores varied considerably even within the same job role—for instance, Data Analyst resumes ranged from 28% to 85%, despite an aggregated average of 47.0% in Table 1—indicating that suitability is driven more by how closely an individual resume's skill set aligns with the job description than by the job role itself. Consistently low scores for roles such as Python Developer (37% in both tested resumes) suggest a persistent skill gap in those resumes rather than a role-specific limitation of the matching algorithm.

The experimental evaluation confirms that the AI-based Resume Analyzer is effective in automating resume screening and providing reliable candidate-job matching results, thereby improving efficiency in the recruitment process.

6. CONCLUSION

The AI-based Resume Analyzer system provides an efficient and automated approach to streamline the recruitment and candidate screening process. By integrating Natural Language Processing (NLP) and Machine Learning techniques, the system successfully extracts relevant information from resumes and matches it with predefined job role requirements.

The proposed system reduces the dependency on manual resume screening, thereby minimizing human effort, bias, and errors in the selection process. It also improves the speed and accuracy of candidate evaluation by generating a structured resume score and classification based on skill relevance and job suitability.

The experimental results demonstrate that the system is capable of effectively handling multiple resumes and providing meaningful insights such as skill extraction, job role recommendation, and candidate ranking. This makes it a valuable tool for enhancing recruitment efficiency in organizations.

Overall, the system achieves its objective of automating resume analysis and improving decision-making in human resource management. It shows strong potential for real-world application in AI-driven recruitment platforms.

7. FUTURE WORK

Future enhancements of this system may include expanding support for multilingual resume processing to handle a wider range of candidates. Additionally, integration with Applicant Tracking Systems (ATS) and cloud-based deployment can improve scalability and real-time recruitment capabilities. Further improvements may also include advanced deep learning models for more accurate semantic understanding of resumes and job descriptions.

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